

# HOW TO UPGRADE YOUR AI AGENT — AND YOUR OWN BRAIN

The Cognitive Neuroscience Playbook for Smarter Autonomous Agents and Sharper, Calmer Humans.

DUAL-MIND BLUEPRINT  
1-PAGE ACTION SHEET

## THE MEMORY LADDER:

1. WORKING (Desk)

→ 2. EPISODIC (Diary)

→ 3. SEMANTIC (Shelf)

→ 4. PROCEDURAL (Autopilot)

Each step down trades detail for 10x execution speed!

## 1. THE DESK: Working Memory & Attention Gating

Baddeley • Cowan (~4 Chunks) • Lost-in-the-Middle

### UPGRADE YOUR BRAIN

- ▶ **Chunk to 3-4 items:** You only hold ~4 chunks. Group chaotic to-dos into 3 bold themes per day.
- ▶ **Clear the clutter:** 50 open browser tabs silently dilute your prefrontal focus. Push them off the desk.
- ▶ **Top-down spotlight:** Aim attention at your goal. Silence bottom-up notification grabs (pagers/pings).

### UPGRADE YOUR AI AGENT

- ▶ **Desk, not warehouse:** Context is not an attic. Strip intermediate logs; cap scratchpad at <2k tokens.
- ▶ **Pin at the poles:** Put the core goal at the start and key rules at the end. Dodge the middle blind spot.
- ▶ **Mask unneeded tools:** Hide irrelevant tool manifests during sub-tasks to stop prompt distraction.

## 2. THE DIARY & THE SHELF: Grounding Facts with Receipts

Tulving • Hippocampus vs Cortex • Recency + Relevance

### UPGRADE YOUR BRAIN

- ▶ **Log surprises immediately:** Human memory drifts and rebuilds. Write 2-line notes right after surprises.
- ▶ **Attach source receipts:** When learning facts, attach who/when/where. A fact without a scene is hearsay.
- ▶ **Combat false memory:** Check original written notes rather than trusting confident intuition.

### UPGRADE YOUR AI AGENT

- ▶ **Log first, summarize later:** Never replace raw run logs with lossy summaries; logs are ground truth.
- ▶ **Vector + Graph receipts:** Require a verifiable citation (source URL + chunk hash) before returning facts.
- ▶ **Retrieve by 3 signals:** Search by *relevance + recency + importance*, never by cosine similarity alone.

## 3. THE SLEEP COMPILER: Turning Practice into Autopilot Habits

Basal Ganglia • Offline Replay • Gated SKILL.md

### UPGRADE YOUR BRAIN

- ▶ **Protect slow-wave sleep:** Your hippocampus replays daytime lessons at night to bake them into habits.
- ▶ **Deliberate practice with feedback:** Autopilot habits are rigid; only conscious critique prevents bad routines.
- ▶ **Spacing over cramming:** Three 30-min spaced sessions consolidate far deeper than one 5-hour marathon.

### UPGRADE YOUR AI AGENT

- ▶ **Consolidate offline behind gates:** Run nightly cron jobs on failure logs to draft candidate SKILL.md runbooks.
- ▶ **Writing habits is high risk:** A wrong skill runs every time! Only compile stable, tested, repeated routines.
- ▶ **Regression test suite:** Rerun 20 known-answer cases before publishing any skill. Zero regressions allowed.

## 4. THE FORK: Fast vs. Slow Mode & Pre-Mortem Rehearsal

Kahneman System 1/2 • Klein Pre-Mortem • Dry-Run Sandbox

### UPGRADE YOUR BRAIN

- ▶ **Pause on obvious hunches:** Catch the "bat-and-ball" trap. When an answer feels fast, engage 5s of System 2.
- ▶ **Run a 60-second Pre-Mortem:** Before launching: "It failed horribly 6 months from now. Why?" Add a guard.
- ▶ **Predict before checking:** Guess the answer before googling. The surprise delta is where learning lives.

### UPGRADE YOUR AI AGENT

- ▶ **Fast by default, slow on purpose:** Route trivial queries to <30ms System 1; route state mutations to System 2.
- ▶ **Dry-run sandbox before writes:** Run destructive actions in shadow mode or plan mode before touching real APIs.
- ▶ **Expectation assertion:** State the expected outcome before calling a tool; abort if observation clashes.

## 5. THE HONEST WATCHER: Calibrated Confidence & Saying "I Don't Know"

Nelson & Narens • Miyake Core Three • Escalation Thresholds

### UPGRADE YOUR BRAIN

- ▶ **Calibrate your ego:** People are notoriously overconfident. Treat a 90% feeling of knowing as a cue to verify.
- ▶ **Reward 'I don't know':** Admitting ignorance unlocks research. Faking certainty leads to catastrophic blunders.
- ▶ **Inhibit impulses (The Pause):** Strengthen the prefrontal muscle: stop automatic reflexes before they speak.

### UPGRADE YOUR AI AGENT

- ▶ **Discount self-reported confidence:** Stated confidence is generated text. Calibrate thresholds from empirical data.
- ▶ **Reward abstention:** Make "I do not know" an allowed answer. If confidence dips or stakes are high, escalate to human.
- ▶ **Ring-0 Inhibitory Gate:** Intercept hallucinations, credential leaks, and raw shell drops at the boundary.

## THE UNIFIED LAW OF COGNITIVE DESIGN:

Treat the context window as a desk (not an attic). Consolidate messy daytime logs into tested nighttime habits. Rehearse irreversible actions in a sandbox. And above all, reward **calibrated uncertainty** over fluent, confident hallucination.

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BRAIN • AGENT